ECONOMETRICA
JOURNAL OF THE ECONOMETRIC SOCIETY

Pseudo Maximum Likelihood Methods: Theory

Author(s): C. Gourieroux, A. Monfort and A. Trognon

Source: *Econometrica*, May, 1984, Vol. 52, No. 3 (May, 1984), pp. 681-700

Published by: The Econometric Society

Stable URL: https://www.jstor.org/stable/1913471

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

The Econometric Society is collaborating with JSTOR to digitize, preserve and extend access to *Econometrica*

JSTOR

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

Econometrica, Vol. 52, No. 3 (May, 1984)

PSEUDO MAXIMUM LIKELIHOOD METHODS: THEORY

BY C. GOURIEROUX, A. MONFORT, AND A. TROGNON

Estimators obtained by maximizing a likelihood function are studied in the case where the true p.d.f. does not necessarily belong to the family chosen for the likelihood function. When such a procedure is applied to the estimation of the parameters of the first order moments, it is possible to prove a necessary and sufficient condition for its consistency. Asymptotic normality is shown as well as the existence of a lower bound for the asymptotic covariance matrix. It is also seen that this bound can be reached if consistent estimates are available for the parameters of the second order moments. Finally, a necessary and sufficient condition for the consistency of the pseudo maximum likelihood estimation of the first and second moments is given.

1. INTRODUCTION

IN THIS PAPER we are interested in the properties of the estimators obtained by maximizing a likelihood function associated with a family of probability distributions, which does not necessarily contain the true distribution. This kind of method will be called the pseudo maximum likelihood method.[1]

More precisely, we determine the families of pseudo likelihood functions providing consistent and asymptotically normal estimators of parameters involved in the true distribution.

After some preliminaries in Section 2, we define in Section 3 the linear exponential families and we show in Section 4 that this kind of family gives consistent and asymptotically normal estimators of the parameters appearing in the first order moment of the true distribution; moreover, it is shown that, conversely, any family with this property is necessarily linear exponential.

In Section 5, we propose a generalization of the previous method adapted to the case in which it is possible to estimate consistently the second order moments of the true distribution. Finally, in Section 6 we define the quadratic exponential families and we show that these families are the only ones providing consistent estimators of the parameters appearing in the first and the second moments. Some technical proofs are gathered in five appendices and the application of the method of scoring is discussed in another one.

2. PRELIMINARIES

The approach used in this paper rests upon previous results by Jennrich [**7**], Malinvaud [**9**], Gallant and Holly [**3**], and Burguete, Gallant, and Souza [**2**] (see also White [**14**]); these results establish the asymptotic properties of estimators obtained by maximizing a function.

[1] Two terminologies, "pseudo" and "quasi", appear in the literature: "pseudo" in the papers on the information criteria (see Sawa [**13**]), "quasi" in some econometric papers (see Hood–Koopmans [**6**]). However "quasi" is ambiguous, since it also refers to "quasi generalized" estimation methods, i.e., to two-step estimation methods, which are also used in this paper.

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

Let us consider G-variate vectors y_t, $t = 1, \ldots, T$, generated by the model:

$$(1) \qquad y_t = f(x_t, \theta_0) + e_t$$

where $\theta_0 \in \Theta \subset \mathbb{R}^k$, $x_t \in \mathbb{R}^p$, $y_t \in \mathbb{R}^G$, $e_t \in \mathbb{R}^G$. We shall assume that the conditional distribution of $e_1, \ldots, e_T$ given $x_1, \ldots, x_T$ is equal to the product of the conditional distributions $\mathscr{L}(e_t \mid x_t)$, where $\mathscr{L}(e_t \mid x_t = x) = \mathscr{L}(e_\tau \mid x_\tau = x)$ for $t \neq \tau$. The true conditional distribution of y_t given x_t, which is unknown, will be denoted by $\lambda_0(x_t, \theta_0)$; it will be assumed that the conditional expectation of y_t is $f(x_t, \theta_0)$ and that its conditional covariance matrix $\Omega_0(x_t)$ exists for any x_t; E_0 will denote the mathematical expectation with respect to $\lambda_0(x, \theta_0)$. We want to estimate θ_0 by considering the solutions of the problem:

$$\operatorname*{Max}_{\theta \in \Theta} \sum_{t=1}^{T} \varphi(y_t, x_t, \theta).$$

Under classical assumptions, denoted by α (see Appendix 1, Section 1), it can be shown that: (a) $\varphi_T(y^T, x^T, \theta) = (1/T)\sum_{t=1}^{T} \varphi(y_t, x_t, \theta)$ converges almost surely, uniformly on Θ, to $\varphi_\infty(\theta) = E_x E_0 \varphi(y, x, \theta)$ where E_x is the expectation with respect to a probability measure μ not depending on φ, θ, and f; (b) $\varphi_\infty(\theta)$ is twice differentiable with respect to θ and the partial derivatives of φ_∞ may be obtained by inverting the expectation and derivative operators; (c) if φ_∞ has a unique maximum at θ_0, then the estimators $\hat{\theta}_T$ obtained by maximizing $\varphi_T(y^T, x^T, \theta)$ exist for all T and converge a.s. to θ_0; moreover $\sqrt{T}(\hat{\theta}_T - \theta_0)$ converges in distribution to $N(0, J^{-1} I J^{-1})$ where

$$J = E_x E_0 \left(- \frac{\partial^2 \varphi(y, x, \theta_0)}{\partial \theta \, \partial \theta'} \right) \quad \text{and}$$

$$I = E_x E_0 \left(\frac{\partial \varphi}{\partial \theta}(y, x, \theta_0) \frac{\partial \varphi}{\partial \theta}(y, x, \theta_0)' \right).$$

These results can be extended to the case where φ also depends on a nuisance parameter, η. If the "nuisance parameter" η is replaced by an estimator $\hat{\eta}_T$ which converges a.s. to some value η_1, and which is such that $\sqrt{T}(\hat{\eta}_T - \eta_1)$ is bounded in probability, then the estimators $\hat{\theta}_T$ obtained by maximizing $(1/T)\sum_{t=1}^{T} \varphi(y_t, x_t, \theta, \hat{\eta}_T)$ have exactly the same properties as before if η is replaced by η_1 in φ_∞, J, and I (see Burguete, Gallant, and Souza [**2**]).

3. LINEAR EXPONENTIAL FAMILIES

In this section we shall use classical results on Fisher information, Kullback information, identification and exponential families; these results may be found in Monfort [**11**].

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

DEFINITION 1: A family of probability measures on $\mathbb{R}^G$, indexed by a parameter $m \in \mathscr{M} \subset \mathbb{R}^G$ is called linear exponential, if: (a) every element of the family has a density function with respect to a given measure $\nu(du)$ and this density function can be written as

$$l(u,m) = \exp\{A(m) + B(u) + C(m)u\}, \qquad u \in \mathbb{R}^G,$$

where $A(m)$ and $B(u)$ are scalar and $C(m)$ is a row vector of size G; (b) m is the mean of the distribution whose density is equal to $l(u,m)$.

Note that (a) and standard properties of Laplace transform imply that all the moments exist. We also assume that conditions α are satisfied for f such that $f(x,\theta) \in \mathscr{M}$, $\forall x$, $\forall \theta$ and for φ defined by: $\varphi(y,x,\theta) = \log l[y, f(x,\theta)]$. This implies, in particular, that the functions A and C are twice continuously differentiable on $\mathring{\mathscr{M}}$, the interior of $\mathscr{M}$.

PROPERTY 1: If $(l(u,m), m \in \mathscr{M})$ is a linear exponential family, we have

$$\frac{\partial A(m)}{\partial m} + \frac{\partial C(m)}{\partial m} m = 0.$$

PROOF: Differentiating the equality

$$\int l(u,m)\nu(du) = 1,$$

we obtain:

$$\int\left(\frac{\partial A(m)}{\partial m} + \frac{\partial C(m)}{\partial m} u\right) l(u,m)\nu(du) = 0 \quad \text{or}$$

$$\frac{\partial A(m)}{\partial m} + \frac{\partial C(m)}{\partial m} m = 0. \qquad Q.E.D.$$

PROPERTY 2:

$$\frac{\partial^2 A}{\partial m\,\partial m'} + \sum_{g=1}^{G} \frac{\partial^2 C_g}{\partial m\,\partial m'} m_g + \frac{\partial C(m)}{\partial m} = 0$$

where C_g and m_g are respectively the gth component of C and m.

PROOF: This property is obtained by differentiating the equality of Property 1. *Q.E.D.*

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

PROPERTY 3: $\partial C/\partial m = \Sigma^{-1}$, where $\Sigma = \int (u-m)(u-m)' l(u,m)\nu(du)$ is the covariance matrix associated with $l(u,m)$.

PROOF: From part (b) of Definition 1, we have

$$\int u l(u,m)\nu(du) = m.$$

According to the analycity property of the Laplace transform we know that we can differentiate the previous equality and that we obtain

$$\int u\left(\frac{\partial A'}{\partial m} + u'\frac{\partial C'}{\partial m}\right) l(u,m)\nu(du) = I_G$$

where I_G is the identity matrix. Using Property 1, we deduce that

$$\int u(u-m)'\frac{\partial C'}{\partial m} l(u,m)\nu(du) = I_G$$

$$\Leftrightarrow \Sigma\frac{\partial C'}{\partial m} = I_G$$

$$\Leftrightarrow \frac{\partial C}{\partial m} = \Sigma^{-1}. \qquad Q.E.D.$$

PROPERTY 4: $\forall m, m_0 \in \mathscr{M}$, we have

$$A(m) + C(m)m_0 \leq A(m_0) + C(m_0)m_0.$$

The equality holds, if and only if $m = m_0$.

PROOF: Kullback's inequality implies

$$\int \log l(u,m) l(u,m_0)\nu(du) \leq \int \log l(u,m_0) l(u,m_0)\nu(du), \quad \text{or}$$

$$A(m) + C(m)m_0 \leq A(m_0) + C(m_0)m_0.$$

The equality is possible if and only if: $l(u,m) = l(u,m_0)$ ν a.s., or, since m is identifiable, if and only if $m = m_0$. *Q.E.D.*

Many classical families of probability measures are linear exponential. Some examples of such families are given in Table I.

The multivariate generalizations of Poisson, negative binomial distributions (Johnson and Kotz [**8**, Chap. 11]) are also linear exponential families.

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

TABLE I

EXAMPLES OF LINEAR EXPONENTIAL FAMILIES

Family	$\mathscr{M}$	Density function	$C(m)$
Binomial (n given)	$]0, n[$	$\frac{\Gamma(n+1)}{\Gamma(u+1)\Gamma(n-u+1)}\left(\frac{m}{n}\right)^u\left(1-\frac{m}{n}\right)^{n-u}$	$\operatorname{Log}\frac{m}{n-m}$
Poisson	$\mathbb{R}^+ - \{0\}$	$\frac{e^{-m}m^u}{u!}$	$\operatorname{Log} m$
Negative binomial (a given)	$\mathbb{R}^+ - \{0\}$	$\frac{\Gamma(a+u)}{\Gamma(a)\Gamma(u+1)}\left(\frac{m}{a}\right)^u\left(1+\frac{m}{a}\right)^{-(a+u)}$	$\operatorname{Log}\frac{m}{a+m}$
Gamma (a given)	$\mathbb{R}^+ - \{0\}$	$\frac{u^{a-1}e^{-ua/m}}{\Gamma(a)\left(\frac{m}{a}\right)^a}$	$-\frac{a}{m}$
Normal (σ given)	$\mathbb{R}$	$\frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{1}{2}\frac{(u-m)^2}{\sigma^2}\right]$	$\frac{m}{\sigma^2}$
Multinomial (n given)	$\sum_g m_g = n$, $m_g > 0$	$\frac{n!}{\prod_g (u_g)!}\prod_g\left(\frac{m_g}{n}\right)^{u_g}$	$\operatorname{Log}\frac{m_1}{n}, \ldots, \operatorname{Log}\frac{m_g}{n}$
Normal multivariate (Σ given)	$\mathbb{R}^G$	$\frac{\exp -\frac{1}{2}(u-m)'\Sigma^{-1}(u-m)}{(2\pi)^{G/2}\sqrt{\det\Sigma}}$	$m'\Sigma^{-1}$

4. PSEUDO MAXIMUM LIKELIHOOD ESTIMATOR BASED ON LINEAR EXPONENTIAL FAMILY

Let us now turn to the problem of estimating θ_0 in model (1): $y_t = f(x_t, \theta_0) + e_t$, defined in Section 2. We assume the first order identification of θ: $f(x, \theta_1) = f(x, \theta_2)$ μ a.s. $\Rightarrow \theta_1 = \theta_2$. We are interested in pseudo maximum likelihood estimators of θ_0, obtained by maximizing $\sum_{t=1}^T \log l[y_t, f(x_t, \theta)]$, where $l(u, m)$ is a family of p.d.f. indexed by the mean $m \in \mathscr{M}$ and where $f(x, \theta) \in \mathscr{M}$, $\forall x$, $\forall \theta$. Conditions $\mathscr{a}$ are assumed to be satisfied for f and φ defined by $\varphi[y, x, \theta] = \log l[y, f(x, \theta)]$.

THEOREM 1: *Under the previous assumptions and if* $(l(u, m), m \in \mathscr{M})$ *is a linear exponential family, a pseudo-maximum likelihood estimator* (PMLE) *of* θ_0 *is strongly consistent.*

PROOF: Using the results mentioned in the preliminaries, the strong consistency will be established, if it is shown that the limit function $\varphi_\infty(\theta) = E_x E_0 \log l[y, f(x, \theta)]$ has a unique maximum at θ_0. It is the case, since $\varphi_\infty(\theta)$ may be written: $\varphi_\infty(\theta) = E_x\{E_0 B(y) + A[f(x, \theta)] + C[f(x, \theta)]f(x, \theta_0)\}$, and the result follows directly from Property 4 and from the first order identifiability of θ. *Q.E.D.*

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

Note that it is equivalent to maximize $\sum_{t=1}^{T}\log l(y_t, f(x_t,\theta))$ or to maximize $\sum_{t=1}^{T}\{A[f(x_t,\theta)] + C[f(x_t,\theta)]y_t\}$. Therefore it is not necessary to impose on y_t the constraints which may be implied by the definition of B. For instance, the pseudo maximum likelihood method with Poisson family may be applied even if y_t is any real number; however, in that case, $f(x_t,\theta)$ must be positive, for any t.

The objective function which is maximized in the PML approach depends on the chosen pseudo family, and it may be interesting to exhibit the objective functions associated with the classical families (see Table II).

TABLE II

OBJECTIVE FUNCTIONS

Family	Objective function
Binomial (n given, $n \in \mathbb{N}$)	$\sum_{t=1}^{T}\left\{n\,\mathrm{Log}\left[1-\frac{f(x_t,\theta)}{n}\right]+y_t\mathrm{Log}\left[\frac{f(x_t,\theta)}{n-f(x_t,\theta)}\right]\right\}$
Poisson	$\sum_{t=1}^{T}\{-f(x_t,\theta)+y_t\mathrm{Log}\,f(x_t,\theta)\}$
Negative Binomial (a given)	$\sum_{t=1}^{T}\left[-a\,\mathrm{Log}\left(1+\frac{f(x_t,\theta)}{a}\right)+y_t\mathrm{Log}\left(\frac{f(x_t,\theta)}{a+f(x_t,\theta)}\right)\right]$
Gamma (a given)	$\sum_{t=1}^{T}\left(-a\,\mathrm{Log}\,f(x_t,\theta)-\frac{ay_t}{f(x_t,\theta)}\right)$
Normal (σ given)	$-\frac{1}{\sigma^2}\sum_{t=1}^{T}[y_t-f(x_t,\theta)]^2$
Multinomial (n given, $n \in \mathbb{N}$)	$\sum_{t=1}^{T}\sum_{g=1}^{G}y_{gt}\mathrm{Log}\,f_g(x_t,\theta)$ where $\sum_g f_g(x_t,\theta)=n$
Normal multivariate (Σ given)	$-\sum_{t=1}^{T}[y_t-f(x_t,\theta)]'\Sigma^{-1}[y_t-f(x_t,\theta)]$

For the univariate and multivariate normal families, the PMLE are, respectively, the nonlinear least squares estimator and a minimum distance estimator (E. Malinvaud [**10**]).

THEOREM 2: *A necessary condition for the* PMLE *associated with a family* $l(u,m)$, $m \in \mathscr{M}$ *(the closure of an open connected set) to be strongly consistent for any* Θ, f, λ_0 *satisfying* α *is that* $l(u,m)$ *be a linear exponential family.*

PROOF: See Appendix 2.

THEOREM 3: *If* $(l(u,m), m \in \mathscr{M})$ *is a linear exponential family, the PMLE* $\hat{\theta}_T$ *is such that*:

$$\sqrt{T}(\hat{\theta}_T-\theta_0)\xrightarrow{d}N(0,J^{-1}IJ^{-1})$$

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

where:

$$J = E_x\left(\frac{\partial f}{\partial \theta}\frac{\partial C}{\partial m}\frac{\partial f'}{\partial \theta}\right) = E_x\left(\frac{\partial f}{\partial \theta}\Sigma_0^{-1}\frac{\partial f'}{\partial \theta}\right),$$

$$I = E_x\left(\frac{\partial f}{\partial \theta}\frac{\partial C}{\partial m}\Omega_0\frac{\partial C'}{\partial m}\frac{\partial f'}{\partial \theta}\right)$$

$$= E_x\left(\frac{\partial f}{\partial \theta}\Sigma_0^{-1}\Omega_0\Sigma_0^{-1}\frac{\partial f'}{\partial \theta}\right).$$

$\Sigma_0 = \Sigma(x,\theta_0)$ [*resp.* $\Omega_0 = \Omega(x,\theta_0)$] *is the covariance matrix associated with the* p.d.f. $l[\cdot, f(x,\theta_0)]$ (*resp. with the distribution* $\lambda_0(x,\theta_0)$) *and* $\partial f/\partial\theta$ *is evaluated at* $\theta = \theta_0$.

PROOF: The asymptotic normality is a consequence of the results mentioned in the preliminaries. The expressions of J and I are derived in Appendix 4.

5. QUASI-GENERALIZED PSEUDO MAXIMUM LIKELIHOOD (QGPML)

PROPERTY 5: The set of asymptotic covariance matrices of the PMLE $\hat{\theta}_T$ based on a linear exponential family has a lower bound equal to

$$\mathscr{K} = \left[E_x\left(\frac{\partial f}{\partial \theta}\Omega_0^{-1}\frac{\partial f'}{\partial \theta}\right)\right]^{-1}.$$

PROOF:

$$J^{-1}IJ^{-1} - \left[E_x\left(\frac{\partial f}{\partial \theta}\Omega_0^{-1}\frac{\partial f'}{\partial \theta}\right)\right]^{-1}$$

$$= E_x\left\{\left[\mathscr{K}\frac{\partial f}{\partial \theta}\Omega_0^{-1}\Sigma_0^{+1/2} - J^{-1}\frac{\partial f}{\partial \theta}\Sigma_0^{-1/2}\right]\Sigma_0^{-1/2}\Omega_0\Sigma_0^{-1/2}\right.$$

$$\left.\times\left[\Sigma_0^{+1/2}\Omega_0^{-1}\frac{\partial f'}{\partial \theta}\mathscr{K} - \Sigma_0^{-1/2}\frac{\partial f'}{\partial \theta}J^{-1}\right]\right\} \gg 0. \qquad Q.E.D.$$

Note that this lower bound is with respect to the PMLE based on linear exponential families and that further efficiency could be gained if it is possible to correctly specify the likelihood function.

As it is shown hereafter the lower bound can be achieved by a two step estimation procedure, when the functional form of the true conditional second order moment of y_t given x_t is known: $V(y_t \mid x_t) = \Omega_0(x_t) = h(x_t, \alpha_0)$.

As shown in Table II, some classical linear exponential families (negative binomial, gamma, normal) also depend on an additional parameter η. This parameter is a differentiable function of m and Σ: $\eta = \Psi(m, \Sigma)$, where Ψ defines,

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

for any given m, a one to one relationship between η and Σ. Therefore it is interesting to consider the more general class of distributions:

$$l^*(u,m,\eta) = \exp\{A(m,\eta) + B(\eta,u) + C(m,\eta)u\}.$$

THEOREM 4: *Let $\tilde{\alpha}_T$ and $\tilde{\theta}_T$ be strongly consistent estimators of α_0 and θ_0, such that $\sqrt{T}(\tilde{\alpha}_T - \alpha_0)$ and $\sqrt{T}(\tilde{\theta}_T - \theta_0)$ are bounded in probability; the* QGPML *estimator of θ, obtained by solving*

$$\max_{\theta} \sum_{t=1}^{T} \log l^*\{y_t, f(x_t,\theta), \Psi[f(x_t,\tilde{\theta}_T), h(x_t,\tilde{\alpha}_T)]\}$$

is, under α, strongly consistent and best asymptotically normal, i.e.:

$$\sqrt{T}(\hat{\theta}_T - \theta_0) \xrightarrow{d} N(0;\mathscr{K}).$$

PROOF: Theorem 4 is a consequence of Burguete, Gallant, and Souza's results, when a nuisance parameter is replaced by a strongly consistent estimator (see Appendix 1, Section 2). The asymptotic covariance matrix of this estimator is the same as the asymptotic covariance matrix of the PML estimator associated with $l(u,m) = l^*(u,m,\Psi[f(x,\theta_0),\Omega_0(x)])$. In this case the matrix Σ, evaluated at $m_0 = f(x,\theta_0)$, is equal to:

$$\frac{\partial C^{-1}}{\partial m}\{f(x,\theta_0), \Psi[f(x,\theta_0),\Omega_0(x)]\} = \Omega_0(x)$$

and the asymptotic matrix is:

$$J^{-1}IJ^{-1} = \left[E_x\left(\frac{\partial f}{\partial\theta}\Omega_0^{-1}\frac{\partial f'}{\partial\theta}\right)\right]^{-1} E_x\left(\frac{\partial f}{\partial\theta}\Omega_0^{-1}\Omega_0\Omega_0^{-1}\frac{\partial f'}{\partial\theta}\right)$$
$$\times\left[E_x\left(\frac{\partial f}{\partial\theta}\Omega_0^{-1}\frac{\partial f'}{\partial\theta}\right)\right]^{-1} = \mathscr{K}. \qquad Q.E.D.$$

As an example let us consider the following model:

$$y_t = f(x_t,\theta_0) + e_t \qquad \text{with} \quad V(y_t \mid x_t) = g^2(x_t,\theta_0)\gamma_0.$$

In a first step, θ_0 may be consistently estimated by the nonlinear least squares estimator $\tilde{\theta}_T$ and a consistent estimate of γ_0 is

$$\tilde{\gamma}_T = \frac{1}{T}\sum_{t=1}^{T}\frac{[y_t - f(x_t,\tilde{\theta}_T)]^2}{g^2(x_t,\tilde{\theta}_T)}.$$

In a second step, it is possible to adopt the QGPML approach using for instance either the normal family or the gamma family. In the first case, the QGPML of θ

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

is obtained by minimizing

$$\sum_{t=1}^{T} \frac{\left[y_t - f(x_t, \theta)\right]^2}{g^2(x_t, \tilde{\theta}_T)\tilde{\gamma}_T},$$

which is simply the quasi generalized nonlinear least squares method. In the second case, the QGPMLE of θ is obtained by maximizing

$$\sum_{t=1}^{T} -\tilde{a}_T(x_t)\left[\log f(x_t, \theta) + \frac{y_t}{f(x_t, \theta)}\right]$$

$$= \sum_{t=1}^{T} -\frac{f^2(x_t, \tilde{\theta}_T)}{g^2(x_t, \tilde{\theta}_T)\tilde{\gamma}_T}\left[\log f(x_t, \theta) + \frac{y_t}{f(x_t, \theta)}\right].$$

These two estimators are asymptotically equivalent and reach the lower bound. In the particular case $g(x_t, \theta) = 1$, the QGPML estimator based on the normal family is equal to the PML estimator. Similarly if $g(x_t, \theta) = f(x_t, \theta)$ the QGPML estimator and the PML estimator based on the gamma family are identical.

Another illustration of Theorem 4 can be found in dichotomous qualitative response models with repeated observations. The observed endogenous variables y_{hi} $(h = 1, \ldots, H;\ i = 1, \ldots, n_h)$ are independent and are such that $P(y_{hi} = 1) = F(x_h\theta)$, $P(y_{hi} = 0) = 1 - F(x_h\theta)$. The properties of the estimators are discussed when $n_h = \lambda_h n$ and $n \to \infty$. The ML estimator of θ, which is a PML estimator based on the Bernoulli distribution, has an asymptotic covariance matrix equal to $\mathscr{K}$. In effect, since it is a PML estimator, its covariance matrix is not less than $\mathscr{K}$; it is equal to $\mathscr{K}$, since this estimator is also asymptotically efficient. The minimum chi-square estimator obtained by minimizing:

$$\sum_{h=1}^{H}\sum_{i=1}^{n_h} \frac{\left[y_{hi} - F(x_h\theta)\right]^2}{\bar{y}_h(1 - \bar{y}_h)}, \qquad \text{where} \quad \bar{y}_h = \frac{1}{n_h}\sum_{i=1}^{n_h} y_{hi}$$

is a QGPLM estimator based on the normal distribution. Therefore this estimator is asymptotically efficient.

A last property of the QGPML estimators is the asymptotic efficiency, when the "true density function" belongs to the family:

$$\lambda^*(y_t, x_t, \theta, \alpha) = \exp\{A^*[m(x_t), \eta(x_t)] + B^*[\eta(x_t), y_t] + C^*[m(x_t), \eta(x_t)]y_t\}$$

with $m(x_t) = f(x_t, \theta)$, $\eta(x_t) = g(x_t, \alpha)$, and θ, α functionally independent. This true density function λ_0^* is obtained for $\theta = \theta_0$ and $\alpha = \alpha_0$.

THEOREM 5: *If assumptions α are satisfied and if the true density function belongs to the previous family the* QGPML *estimator of θ_0 is asymptotically equivalent to the maximum likelihood estimator of θ_0 obtained by maximizing (with respect to θ and α):* $\sum_{t=1}^{T}\log\lambda^*(y_t, x_t, \theta, \alpha)$.

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

PROOF: Since the ML estimators

$$\begin{pmatrix} \theta_T^* \\ \alpha_T^* \end{pmatrix}$$

of the parameters are consistent and asymptotically normal, we only have to compare $\mathscr{K}$ with the asymptotic covariance matrix of θ_T^*. This matrix is obtained from:

$$V_{\text{as}}\sqrt{T}\begin{pmatrix} \theta_T^* - \theta_0 \\ \alpha_T^* - \alpha_0 \end{pmatrix} = \left[E_x E_0\left(- \frac{\partial^2 \log \lambda^*(y, x, \theta_0, \alpha_0)}{\partial(\theta, \alpha)\, \partial(\theta, \alpha)'} \right) \right]^{-1}.$$

Let us now show that θ_T^* and α_T^* are asymptotically independent; we have (for notation convenience, we consider the case $G = 1$)

$$\begin{aligned} & E_x E_0\left(- \frac{\partial^2 \log \lambda^*(y, x, \theta_0, \alpha_0)}{\partial\theta\, \partial\alpha'} \right) \\ & \quad = E_x\left(\frac{\partial f}{\partial\theta} E_0\left(\frac{\partial A^*}{\partial m\, \partial\eta'} + y \frac{\partial C^*}{\partial m\, \partial\eta'} \right) \frac{\partial g'}{\partial\alpha} \right) \\ & \quad = E_x\left[\frac{\partial f}{\partial\theta} \left(\frac{\partial A^*}{\partial m\, \partial\eta'} + m_0 \frac{\partial C^*}{\partial m\, \partial\eta'} \right) \frac{\partial g'}{\partial\alpha} \right] \end{aligned}$$

where all the expressions are evaluated at the true values θ_0, α_0. It is deduced from Property 1 that

$$\frac{\partial A^*}{\partial m\, \partial\eta'} + m_0 \frac{\partial C^*}{\partial m\, \partial\eta'} = 0$$

and the asymptotic independence follows. Therefore:

$$\begin{aligned} & V_{\text{as}}\left[\sqrt{T}(\theta_T^* - \theta_0) \right] \\ & \quad = \left(E_x E_0\left[- \frac{\partial^2 \log \lambda^*(y, x, \theta_0, \alpha_0)}{\partial\theta\, \partial\theta'} \right] \right)^{-1} \\ & \quad = \left(E_x E_0\left(\frac{\partial \log \lambda^*}{\partial\theta} \frac{\partial \log \lambda^{*\prime}}{\partial\theta} \right) \right)^{-1} \\ & \quad = \left(E_x\left[\frac{\partial f}{\partial\theta} E_0\left(\frac{\partial C^*}{\partial m}(y - m)(y - m)' \frac{\partial C^{*\prime}}{\partial m} \right) \frac{\partial f'}{\partial\theta} \right] \right)^{-1} \\ & \quad = \left(E_x\left[\frac{\partial f}{\partial\theta} \Omega_0^{-1} E_0(y - m)(y - m')\Omega_0^{-1} \frac{\partial f'}{\partial\theta} \right] \right)^{-1} \\ & \quad = \left(E_x\left[\frac{\partial f}{\partial\theta} \Omega_0^{-1} \frac{\partial f'}{\partial\theta} \right] \right)^{-1} = \mathscr{K}. \end{aligned}$$

Q.E.D.

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

6. QUADRATIC EXPONENTIAL FAMILIES

The approach of Sections 3 and 4 may be generalized to take into account the second order moments.

The model now considered is the following:

$$y_t = f(x_t, \theta_0) + e_t, \qquad \text{where} \quad V(e_t \mid x_t) = \Omega_0(x_t) = g(x_t, \theta_0).$$

The parameter θ is assumed to be second order identifiable:

$$\begin{cases} f(x,\theta_1) = f(x,\theta_0) \\ g(x,\theta_1) = g(x,\theta_0), \end{cases} \qquad \mu \text{ a.e.} \Rightarrow \theta_0 = \theta_1.$$

The other assumptions of Section 2 are satisfied.

DEFINITION 2: A family of probability measures on $\mathbb{R}^G$ indexed by $m \in \mathscr{M} \subset \mathbb{R}^G$ and $\Sigma \in \mathscr{E}$, where $\mathscr{E}$ is a subset of the positive definite matrices, is called quadratic exponential, if: (a) every element of the family has a density function with respect to a given measure $\nu(du)$, which can be written as:

$$l(u, m, \Sigma) = \exp\{A(m,\Sigma) + B(u) + C(m,\Sigma)u + u'D(m,\Sigma)u\}$$

where $A(m,\Sigma)$, $B(u)$ are scalar, $C(m,\Sigma)$ is a row vector of size G, and $D(m,\Sigma)$ is a square matrix (G,G); (b) m is the mean and Σ is the covariance matrix of the distribution $l(u,m,\Sigma)$.

We also assume that conditions α are satisfied for f, g such that $f(x,\theta) \in \mathscr{M}$, $\forall x$, $\forall \theta$, $g(x,\theta) \in \mathscr{E}$, $\forall x$, $\forall \theta$ and for φ defined by:

$$\varphi(y,x,\theta) = \log l\big[y, f(x,\theta), g(x,\theta)\big].$$

Examples of such families are, for instance, the normal distribution $N(m,\Sigma)$ or the discrete distribution on $\{-1,0,1\}$, with probabilities p_1, p_2, p_3; in the latter case $l(u,m,\sigma^2) = p_1^{u(1+u)/2} p_2^{1-u^2} p_3^{u(-1+u)/2}$ with $p_1 + p_2 + p_3 = 1$, $p_1 - p_3 = m$, $p_1 + p_3 = \sigma^2 + m^2$.

PROPERTY 6: $\forall m, m_0 \in \mathscr{M}$, $\forall \Sigma$, $\Omega_0 \in \mathscr{E}$:

$$A(m,\Sigma) + C(m,\Sigma)m_0 + \operatorname{tr} D(m,\Sigma)(\Omega_0 + m_0 m_0')$$
$$\leq A(m_0,\Omega_0) + C(m_0,\Omega_0)m_0 + \operatorname{tr} D(m_0,\Omega_0)(\Omega_0 + m_0 m_0').$$

The equality is possible, if and only if $m = m_0$, $\Sigma = \Omega_0$.

PROOF: This result is, as for Property 4, a direct consequence of Kullback's inequality. *Q.E.D.*

From this property and the second order identifiability condition, we deduce, as for Theorems 1 and 2, the strong consistency and the asymptotic normality of the PML estimator of θ based on $l(u,m,\Sigma)$.

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

THEOREM 6: *If* $(l(u,m,\Sigma),\ m \in \mathscr{M},\ \Sigma \in \mathscr{E})$ *is a quadratic exponential family, the estimator of* θ *obtained by maximizing*:

$$\max_{\theta} \sum_{t=1}^{T} \log l[\, y_t, f(x_t,\theta), g(x_t,\theta)]$$

is strongly consistent and asymptotically normal.

The proof is similar to that of Theorem 1. Note, however that the expression of the asymptotic covariance matrix is more complicated (see Appendix 5).

It is also possible to obtain the reciprocal of the previous theorem, in the case $G = 1$.

THEOREM 7: *A necessary condition for the* PML *estimator associated with a family* $l(u,m,\sigma^2)$, $m \in \mathscr{M}$, $\sigma^2 \in \mathscr{E}$ ($\mathscr{M}$ *and* $\mathscr{E}$ *are closures of open connected sets*) *to be strongly consistent for any* Θ, f, g, λ_0 *satisfying* α *is that* $l(u,m,\sigma^2)$ *be a quadratic exponential family.*

PROOF: See Appendix 3.

7. CONCLUSION

The results obtained in this paper are clearly related to the general problem of robustness and could be used in many contexts. This is the reason of the high level of generality kept throughout the paper. A companion paper (Gourieroux, Monfort, and Trognon [**5**]) describes the application of these estimation methods to econometric models for discrete data. It is also worth noting that the PML procedure can be extended to some cases where the disturbances are serially correlated (Gourieroux, Monfort, and Trognon [**4**], Robinson [**12**]).

CEPREMAP, Paris, France
and
Ecole Nationale de la Statistique et de l'Administration Economique, Malakoff, France

Manuscript received January, 1982; final revision received May, 1983.

APPENDIX 1

1. ASSUMPTIONS α

These assumptions are deduced directly from the assumptions of Burguete, Gallant, and Souza [**2**], but they have been adapted to our particular case. These assumptions are: (i) f and φ are continuous with respect to all the variables and are twice continuously differentiable with respect to θ; (ii) Θ is a compact set and θ_0 is in the interior of Θ; (iii) almost every realization of (e_t, x_t) is a Cesaro sum generator with respect to the measure:

$$\nu(A) = \int_{\mathbb{R}^P}\int_{\mathbb{R}^G} \mathbb{1}_A(e,x)\, d\mathscr{L}(e/x)\, d\mu(x)$$

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

and to a dominating function $b(e,x)$; (iv) the sequence (x_t) is a Cesaro sum generator with respect to μ and $b(x) = \int_{\mathbb{R}^G} b(e,x)\, d\mathscr{L}(e/x)$; (v) for each $x \in \mathbb{R}^p$, there is a neighborhood N_x such that

$$\int_{\mathbb{R}^G} \sup_{N_x} b(e,x)\, d\mathscr{L}(e/x) < \infty.$$

(vi) functions $\varphi(x, y, \theta)$, $\partial\varphi(y,x,\theta)/\partial\theta_i$, $\partial^2\varphi(y,x,\theta)/\partial\theta_i\,\partial\theta_j$ are dominated by $b(e,x)$.

2. VERIFICATION OF THE BURGUETE–GALLANT–SOUZA θ ASSUMPTIONS FOR THE CASE OF A NUISANCE PARAMETER

The only condition which is not obviously satisfied is

$$\Delta = E_x E\left(\frac{\partial^2}{\partial\theta\,\partial(\tilde{\theta},\tilde{\alpha})} \log l^*[y, f(x,\theta), \Psi[f(x,\tilde{\theta}), h(x,\tilde{\alpha})]]\right)_{\theta=\tilde{\theta}=\theta_0,\ \alpha=\tilde{\alpha}_0} = 0.$$

This equality is true, since

$$\Delta_{ij} = E_x\left\{\left(\frac{\partial f}{\partial\theta_i}\right)'_{\theta_0} E\left(\frac{\partial^2 \log l^*[y, f(x,\theta_0), \Psi[f(x,\theta_0), h(x,\alpha_0)]]}{\partial m\,\partial\eta'}\right)\left(\frac{\partial\Psi}{\partial(\tilde{\theta},\tilde{\alpha})_j}\right)_{\theta_0,\alpha_0}\right\}$$

$$= E_x\left\{\left(\frac{\partial f}{\partial\theta_i}\right)'_{\theta_0} E\left[\sum_g \frac{\partial^2 C_g}{\partial m\,\partial\eta'}(y_g - m_g)\right]\left[\frac{\partial\Psi}{\partial(\tilde{\theta},\tilde{\alpha})_j}\right]_{\theta_0,\alpha_0}\right\}$$

where $m_g = f_g(x,\theta_0)$ and C_g is evaluated at $f(x,\theta_0), \Psi[f(x,\theta_0), h(x,\theta_0)]$. The result follows from $E(y_g - m_g) = 0$.

APPENDIX 2

PROOF OF THEOREM 2: We are going to show that the condition is already necessary, if we consider the special case in which the y_t are i.i.d., with mean $\theta_0 \in \Theta = \mathscr{M} \subset \mathbb{R}^G$.

FIRST STEP: The PMLE exists since Θ is compact and φ_T is continuous on Θ. If this estimator $\hat{\theta}_T$ is strongly consistent for θ_0, we deduce from the strong uniform convergence of $\varphi_T(y^T, x^T, \theta)$ to $\varphi_\infty(\theta)$, that θ_0 provides the maximum of φ_∞. Since $\theta_0 \in \mathring{\Theta}$ and since φ_∞ is differentiable, we have:

$$\frac{d\varphi_\infty}{d\theta}(\theta_0) = E_x E_0 \frac{\partial \log l(y,\theta_0)}{\partial\theta} = E_0 \frac{\partial \log l(y,\theta_0)}{\partial\theta} = 0.$$

SECOND STEP: Let us first consider the case $G = 1$. For any distribution λ_0 such that $E_0 y = \theta_0$ we have

$$E_0 \frac{\partial \log l(y,\theta_0)}{\partial\theta} = 0.$$

In particular, this is true if the support of λ_0 consists in two points y_1 and y_2, with $y_1 < \theta_0 < y_2$; therefore:

$$\forall p_1, p_2 \quad \begin{cases} p_1 + p_2 = 1 \\ p_1 y_1 + p_2 y_2 = \theta_0 \end{cases} \Rightarrow p_1 \frac{\partial \log l(y_1,\theta_0)}{\partial\theta} + p_2 \frac{\partial \log l(y_2,\theta_0)}{\partial\theta} = 0.$$

This implies that for any y_1, y_2 such that $y_1 < \theta_0 < y_2$

$$(y_2 - \theta_0)\frac{\partial \log l(y_1,\theta_0)}{\partial\theta} + (\theta_0 - y_1)\frac{\partial \log l(y_2,\theta_0)}{\partial\theta} = 0,$$

$$\frac{\dfrac{\partial \log l(y_1,\theta_0)}{\partial\theta}}{y_1 - \theta_0} = \frac{\dfrac{\partial \log l(y_2,\theta_0)}{\partial\theta}}{y_2 - \theta_0}.$$

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

Considering y_2 as a variable and y_1 as fixed and then allowing y_1 to vary, it is seen that there exists a scalar $\lambda(\theta_0)$ such that

$$\frac{\partial \log l(y, \theta_0)}{\partial \theta} = \lambda(\theta_0)(y - \theta_0).$$

The result follows by integrating both sides with respect to θ in $\mathring{\mathscr{M}} = \mathring{\Theta}$ which is a connected set and extending to Θ by continuity arguments (see α). In the general G-dimensional case, the same kind of proof can be developed by considering distributions whose supports consist of $G + 1$ independent points admitting θ_0 as a barycenter.

The second step is very similar to Farkas' Lemma. In effect we have proved that:

$$\left\{ \forall \lambda_0 : \int (y - \theta_0)\, d\lambda_0(y) = 0 \Rightarrow \int \frac{\partial \log l(y, \theta_0)}{\partial \theta}\, d\lambda_0(y) = 0 \right\}$$

implies the existence of a linear relationship:

$$\frac{\partial \log \lambda(y, \theta_0)}{\partial \theta} = \lambda(\theta_0)[y - \theta_0].$$

However it is easily seen that the usual assumptions of Farkas' Lemma (Berge [1]) are not satisfied in our case.

APPENDIX 3

PROOF OF THEOREM 7 (Case $G = 1$): We shall see that the condition of Theorem 7 is already necessary if we restrict ourselves to the case in which the y_t are i.i.d. with mean $f(x_t, \theta_0) = m_0$ and variance $g(x_t, \theta_0) = \sigma_0^2$. The same arguments as those given in the first step of Appendix 1 imply that

$$E_0 \frac{\partial \log l(y, m_0, \sigma_0^2)}{\partial m} = 0$$

and

$$E_0 \frac{\partial \log l(y, m_0, \sigma_0^2)}{\partial \sigma^2} = 0.$$

In particular, this is true if the support of λ_0 consists of three points $y_1 < y_2 = m_0 < y_3$; therefore

$$\text{(A3.1)} \qquad \forall p_1, p_2, p_3: \quad \begin{cases} p_1 + p_2 + p_3 = 1, \\ p_1 y_1 + p_2 m_0 + p_3 y_3 = m_0, \\ p_1 y_1^2 + p_2 m_0^2 + p_3 y_3^2 = m_0^2 + \sigma_0^2; \end{cases}$$

$$\text{(A3.2)} \qquad \begin{cases} p_1 \dfrac{\partial \log l(y_1, m_0, \sigma_0^2)}{\partial m} + p_2 \dfrac{\partial \log l(m_0, m_0, \sigma_0^2)}{\partial m} + p_3 \dfrac{\partial \log l(y_3, m_0, \sigma_0^2)}{\partial m} = 0, \\ p_1 \dfrac{\partial \log l(y_1, m_0, \sigma_0^2)}{\partial \sigma^2} + p_2 \dfrac{\partial \log l(m_0, m_0, \sigma_0^2)}{\partial \sigma^2} + p_3 \dfrac{\partial \log l(y_3, m_0, \sigma_0^2)}{\partial \sigma^2} = 0. \end{cases}$$

Denoting $(y_j - m_0)/\sigma_0^2$ by y_j^*, system (A3.1) becomes

$$\begin{cases} p_1 + p_2 + p_3 = 1, \\ p_1 y_1^* + p_3 y_3^* = 0, \\ p_1 y_1^{*2} + p_3 y_3^{*2} = 1, \end{cases} \qquad \text{with} \quad y_1^* < 0 < y_3^*.$$

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

Solving this system, we obtain

$$p_1 = \frac{-1}{y_1^*(y_3^* - y_1^*)}, \qquad p_2 = \frac{y_1^* y_3^* + 1}{y_1^* y_3^*}, \qquad p_3 = \frac{-1}{y_3^*(y_1^* - y_3^*)}.$$

This solution is a probability distribution if, and only if, $y_1^* y_3^* \leqq -1$. For such a probability distribution we have:

$$\begin{cases} p_1 y_1^* + p_3 y_3^* = 0, \\ p_1(y_1^{*2} - 1) - p_2 + p_3(y_3^{*2} - 1) = 0, \\ p_1 \dfrac{\partial \log l^*(y_1^*, m_0, \sigma_0^2)}{\partial m} + p_2 \dfrac{\partial \log l^*(y_2^*, m_0, \sigma_0^2)}{\partial m} + p_3 \dfrac{\partial \log l^*(y_3^*, m_0, \sigma_0^2)}{\partial m} = 0 \end{cases}$$

with

$$\frac{\partial \log l^*}{\partial m}(y_i^*, m_0, \sigma_0^2) = \frac{\partial \log l}{\partial m}(y_i, m_0, \sigma_0^2).$$

Since this homogeneous system has a nonzero solution p_1, p_2, p_3, this implies

$$\Delta(y_1^*, y_3^*) = \det \begin{vmatrix} y_1^* & 0 & y_3^* \\ y_1^{*2} - 1 & -1 & y_3^{*2} - 1 \\ \dfrac{\partial \log l^*(y_1^*, m_0, \sigma_0^2)}{\partial m} & \dfrac{\partial \log l^*(0, m_0, \sigma_0^2)}{\partial m} & \dfrac{\partial \log l^*(y_3^*, m_0, \sigma_0^2)}{\partial m} \end{vmatrix} = 0$$

$\forall y_1^*, y_3^* \in \mathscr{D} = \{y_1^* < 0 < y_3^*, y_1^* y_3^* \leq -1\}$.

This also implies that, for any (y_1^*, y_3^*) belonging to $\mathscr{D}$, the functions

$$\frac{\partial \log l^*}{\partial m}(y_1^*, m_0, \sigma_0^2) \quad \text{and} \quad \frac{\partial \log l^*}{\partial m}(y_3^*, m_0, \sigma_0^2)$$

can be written as:

$$\begin{cases} \dfrac{\partial \log l^*(y_3^*, m_0, \sigma_0^2)}{\partial m} = \alpha(y_1^*) + \beta(y_1^*) y_3^* - \alpha(y_1^*) y_3^{*2}, \\ \dfrac{\partial \log l^*(y_1^*, m_0, \sigma_0^2)}{\partial m} = \tilde{\alpha}(y_3^*) + \tilde{\beta}(y_3^*) y_1^* - \tilde{\alpha}(y_3^*) y_1^{*2}. \end{cases}$$

Considering successively y_1^* as a variable and y_3^* fixed and then y_3^* as a variable and y_1^* fixed, it is seen that the functions, α, $\beta, \tilde{\alpha}$, $\tilde{\beta}$ are constant; therefore we have

$$\begin{cases} \dfrac{\partial \log l^*}{\partial m}(y_3^*, m_0, \sigma_0^2) = \alpha + \beta y_3^* - \alpha y_3^{*2}, & \forall y_3^* > 0, \\ \dfrac{\partial \log l^*(y_1^*, m_0, \sigma_0^2)}{\partial m} = \tilde{\alpha} + \tilde{\beta} y_1^* - \tilde{\alpha} y_1^{*2}, & \forall y_1^* < 0. \end{cases}$$

Replacing in the equation $\Delta(y_1^*, y_3^*) = 0$, $\partial \log l^*(y_3^*, m_0, \sigma_0^2)/\partial m$ and $\partial \log l^*(y_1^*, m_0, \sigma_0^2)/\partial m$ by the previous expressions we obtain

$$\frac{\partial \log l^*(0, m_0, \sigma_0^2)}{\partial m} = \alpha = \tilde{\alpha}$$

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

and $\beta = \tilde{\beta}$. It follows that

$$\frac{\partial \log l^*(y^*, m, \sigma^2)}{\partial m} = \alpha(m, \sigma^2) + \beta(m, \sigma^2) y^* - \alpha(m, \sigma^2) y^{*2}, \qquad \forall m, \sigma^2, y^*,$$

$$\frac{\partial \log l(y, m, \sigma^2)}{\partial m} = a(m, \sigma^2) + c(m, \sigma^2) y + d(m, \sigma^2) y^2, \qquad \forall m, \sigma^2, y.$$

Integrating with respect to m, we get:

$$\text{(A3.3)} \qquad \log l(y, m, \sigma^2) = A(m, \sigma^2) + B(\sigma^2, y) + C(m, \sigma^2) y + D(m, \sigma^2) y^2.$$

The same argument applied to σ^2 instead of m gives:

$$\text{(A3.4)} \qquad \log l(y, m, \sigma^2) = A^*(m, \sigma^2) + B^*(m, y) + C^*(m, \sigma^2) y + D^*(m, \sigma^2) y^2.$$

The previous equation shows that the terms depending simultaneously on y and σ^2 are quadratic in y, and from (A3.3) we obtain

$$\log l(y, m, \sigma^2) = A(m, \sigma^2) + B(y) + C(m, \sigma^2) y + D(m, \sigma^2) y^2.$$

APPENDIX 4

COMPUTATION OF THE ASYMPTOTIC COVARIANCE MATRIX OF THE PSEUDO MAXIMUM LIKELIHOOD ESTIMATOR IN THE CASE OF THE LINEAR EXPONENTIAL FAMILY

As it is proved by Burguete, Gallant, and Souza [2], the asymptotic variance-covariance matrix of the PMLE is given by

$$V = J^{-1} I J^{-1}$$

where

$$J = E_x E_0 \left[-\frac{\partial^2 \log l}{\partial \theta \, \partial \theta'} (y, f(x, \theta_0)) \right],$$

$$I = E_x E_0 \left[\frac{\partial \log l}{\partial \theta} (y, f(x, \theta_0)) \frac{\partial \log l}{\partial \theta} (y, f(x, \theta_0))' \right].$$

If l is an element of a linear exponential family, we have:

$$\text{(A4.1)} \qquad \frac{\partial \log l}{\partial \theta}(y, x, \theta) = \frac{\partial}{\partial \theta} [A(f(x, \theta)) + B(y) + C(f(x, \theta)) y]$$

$$= \frac{\partial f(x, \theta)}{\partial \theta} \left(\frac{\partial A}{\partial m} (f(x, \theta)) + \frac{\partial C}{\partial m} (f(x, \theta)) y \right)$$

$$= \frac{\partial f}{\partial \theta} \left(\frac{\partial A}{\partial m} + \frac{\partial C}{\partial m} y \right)$$

$$= \frac{\partial f}{\partial \theta} \frac{\partial C}{\partial m} (y - m) \qquad \text{(property 1)}$$

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

and

$$\text{(A4.2)} \qquad \frac{\partial^2 \log l}{\partial\theta\,\partial\theta'}(y, f(x,\theta)) = \frac{\partial}{\partial\theta}\left[\sum_g \frac{\partial f_g}{\partial\theta}\left(\frac{\partial A}{\partial m_g} + \frac{\partial C}{\partial m_g}\, y\right)\right]'$$

$$= \sum_g \frac{\partial^2 f_g}{\partial\theta\,\partial\theta'}\left(\frac{\partial A}{\partial m_g} + \frac{\partial C}{\partial m_g}\, y\right) + \frac{\partial f}{\partial\theta}\left[\frac{\partial^2 A}{\partial m\,\partial m'} + \sum_g \frac{\partial^2 C_g}{\partial m\,\partial m'}\, y_g\right]\frac{\partial f'}{\partial\theta}$$

$$= \sum_g \frac{\partial^2 f_g}{\partial\theta\,\partial\theta'}\,\frac{\partial C}{\partial m_g}(y - m) + \frac{\partial f}{\partial\theta}\left[\sum_g \frac{\partial^2 C_g}{\partial m\,\partial m'}(y_g - m_g) - \frac{\partial C}{\partial m}\right]\frac{\partial f'}{\partial\theta}$$

(properties 1 and 2).

From (A4.1) we deduce:

$$I = E_x\left[\frac{\partial f}{\partial\theta}\,\frac{\partial C}{\partial m}\,E_0((y-m)(y-m)')\,\frac{\partial C'}{\partial m}\,\frac{\partial f'}{\partial\theta}\right] = E_x\left[\frac{\partial f}{\partial\theta}\,\frac{\partial C}{\partial m}\,\Omega_0\,\frac{\partial C'}{\partial m}\,\frac{\partial f'}{\partial\theta}\right].$$

From (A4.2) it follows

$$J = -E_x\left\{\frac{\partial f}{\partial\theta}\left[\sum_g \frac{\partial^2 C_g}{\partial m\,\partial m'}\,E_0(y_g - f_g(x,\theta_0)) - \frac{\partial C}{\partial m}\right]\frac{\partial f'}{\partial\theta}\right\}$$

$$= +E_x\left[\frac{\partial f}{\partial\theta}\,\frac{\partial C}{\partial m}\,\frac{\partial f'}{\partial\theta}\right]$$

where $\partial f/\partial\theta, \partial C/\partial m$ stand for $\partial f(x,\theta_0)/\partial\theta, \partial C(f(x,\theta_0))/\partial m$ respectively.

APPENDIX 5

COMPUTATION OF THE ASYMPTOTIC COVARIANCE MATRIX OF THE PSEUDO MAXIMUM LIKELIHOOD ESTIMATOR IN THE CASE OF QUADRATIC EXPONENTIAL FAMILIES ($G = 1$)

Consider the following quadratic exponential family:

$$l(u, m, \sigma^2) = \exp[A(m,\sigma^2) + B(u) + C(m,\sigma^2)u + D(m,\sigma^2)u^2] \qquad (u \in \mathbb{R}),$$

where l is a density function. Then $\int l(u,m,\sigma^2)\,\nu(du) = 1$. Differentiating this last equality with respect to $\alpha' = (m, \sigma^2)$ leads to

$$\text{(A5.1)} \qquad \frac{\partial A}{\partial\alpha} + \frac{\partial C}{\partial\alpha}\,m + \frac{\partial D}{\partial\alpha}(m^2 + \sigma^2) = 0.$$

Since the mean and the variance of l are m and σ^2, it follows:

$$\text{(A5.2)} \qquad \int u l(u,m,\sigma^2)\,\nu(du) = m,$$

$$\text{(A5.3)} \qquad \int u^2 l(u,m,\sigma^2)\,\nu(du) = m^2 + \sigma^2.$$

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

The differentiation of (A5.2) with respect to $\alpha' = (m, \sigma^2)$ implies

$$\text{(A5.4)} \qquad \frac{\partial A}{\partial \alpha} m + \frac{\partial C}{\partial \alpha}(m^2 + \sigma^2) + \frac{\partial D}{\partial \alpha} Eu^3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The differentiation of (A5.3) with respect to α yields

$$\text{(A5.5)} \qquad \frac{\partial A}{\partial \alpha}(m^2 + \sigma^2) + \frac{\partial C}{\partial \alpha} Eu^3 + \frac{\partial D}{\partial \alpha} Eu^4 = \begin{pmatrix} 2m \\ 1 \end{pmatrix}.$$

The computation of the asymptotic covariance matrix of the PMLE is based on

$$J = E_x E_0 \left[-\frac{\partial^2 \log l}{\partial \theta \, \partial \theta'}(y, f(x, \theta_0), g(x, \theta_0)) \right] \quad \text{and}$$

$$I = E_x E_0 \frac{\partial \log l}{\partial \theta}(y, f(x, \theta_0), g(x, \theta_0)) \frac{\partial \log l}{\partial \theta}(y, f(x, \theta_0), g(x, \theta_0))'.$$

The first and second derivatives of $\log l$ are in this case

$$\frac{\partial \log l}{\partial \theta} = \frac{\partial h}{\partial \theta}\left(\frac{\partial A}{\partial \alpha} + \frac{\partial C}{\partial \alpha} y + \frac{\partial D}{\partial \alpha} y^2 \right) \qquad \text{with} \quad h = (f, g),$$

$$\frac{\partial^2 \log l}{\partial \theta \, \partial \theta'} = \frac{\partial h}{\partial \theta}\left(\frac{\partial^2 A}{\partial \alpha \, \partial \alpha'} + \frac{\partial^2 C}{\partial \alpha \, \partial \alpha'} y + \frac{\partial^2 D}{\partial \alpha \, \partial \alpha'} y^2 \right) \frac{\partial h'}{\partial \theta}.$$

Let us first consider the computation of J:

$$J = -E_x E_0 \frac{\partial^2 \log l}{\partial \theta \, \partial \theta'}$$

$$= -E_x \left[\frac{\partial h}{\partial \theta}\left(\frac{\partial^2 A}{\partial \alpha \, \partial \alpha'} + \frac{\partial^2 C}{\partial \alpha \, \partial \alpha'} f(x, \theta_0) + \frac{\partial^2 D}{\partial \alpha \, \partial \alpha'}\left(f(x, \theta_0)^2 + g(x, \theta_0) \right) \right) \frac{\partial h}{\partial \theta'} \right].$$

For expository purpose we will denote $f(x, \theta_0)$ by m_0 and $g(x, \theta_0)$ by σ_0^2. With these notations (A5.1) implies

$$-\left\{ \frac{\partial^2 A}{\partial \alpha \, \partial \alpha'} + \frac{\partial^2 C}{\partial \alpha \, \partial \alpha'} m_0 + \frac{\partial^2 D}{\partial \alpha \, \partial \alpha'}(m_0^2 + \sigma_0^2) \right\} = \frac{\partial C}{\partial \alpha}(1 : 0) + \frac{\partial D}{\partial \alpha}(2m_0 : 1).$$

Finally J is defined by:

$$J = E_x \left\{ \frac{\partial h}{\partial \theta}\left[\frac{\partial C}{\partial \alpha}(1 : 0) + \frac{\partial D}{\partial \alpha}(2m_0 : 1) \right] \frac{\partial h'}{\partial \theta} \right\}$$

where $\partial C / \partial \alpha, \partial D / \partial \alpha$ are computed at $m_0 = f(x, \theta_0)$ and $\sigma_0^2 = g(x, \theta_0)$ and $\partial h / \partial \theta$ at θ_0.

Let us now consider the computation of I which is defined by:

$$I = E_x E_0 \left\{ \frac{\partial \log l}{\partial \theta}(y, f(x, \theta_0), g(x, \theta_0)) \frac{\partial \log l}{\partial \theta}(y, f(x, \theta_0), g(x, \theta_0))' \right\}.$$

Let $m_3^0 = E_0 y^3$ and $m_4^0 = E_0 y^4$ be the third and fourth product moments of λ_0, the true distribution of the observations, and m_3 and m_4 the similar moments for the family considered, evaluated at $m = m_0$, $\sigma^2 = \sigma_0^2$.

$$I = E_x \left\{ \frac{\partial h}{\partial \theta}\left(\frac{\partial C}{\partial \alpha}\left\{ \frac{\partial A}{\partial \alpha} m_0 + \frac{\partial C}{\partial \alpha}(m_0^2 + \sigma_0^2) + \frac{\partial D}{\partial \alpha} m_3^0 \right\}' \right.\right.$$

$$\left.\left. + \frac{\partial D}{\partial \alpha}\left\{ \frac{\partial A}{\partial \alpha}(m_0 + \sigma_0^2) + \frac{\partial C}{\partial \alpha} m_3^0 + \frac{\partial D}{\partial \alpha} m_4^0 \right\}' \right) \frac{\partial h'}{\partial \theta} \right\}.$$

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

From (A5.4) and (A5.5) it follows:

$$I = E_x \left\{ \frac{\partial h}{\partial \theta} \left(\frac{\partial C}{\partial \alpha} \left[(1:0) + \frac{\partial D'}{\partial \alpha} (m_3^0 - m_3) \right] \right.\right.$$

$$\left.\left. + \frac{\partial D}{\partial \alpha} \left[(2m_0 : 1) + \frac{\partial C'}{\partial \alpha} (m_3^0 - m_3) + \frac{\partial D'}{\partial \alpha} (m_4^0 - m_4) \right] \right) \frac{\partial h'}{\partial \theta} \right\}.$$

APPENDIX 6

APPLICATION OF THE METHOD OF SCORING TO LINEAR EXPONENTIAL FAMILIES

Let us examine the PML estimation problem with the following pseudo likelihood function (case $G = 1$):

$$\sum_{t=1}^{T} \log l(y_t, f(x_t b)) = \sum_{t=1}^{T} \{ A(f(x_t b)) + B(y_t) + C(f(x_t b)) y_t \}$$

where $A(\cdot)$, $B(\cdot)$, and $C(\cdot)$ are scalar functions. The pseudo maximum likelihood estimate may be obtained by the method of scoring; the $(h+1)$th iteration is

$$\tilde{b}_{h+1} = \tilde{b}_h - \left[E_0 \sum_{t=1}^{T} \frac{\partial^2 \log l}{\partial b \, \partial b'} \right]^{-1}_{b = \tilde{b}_h} \left[\sum_{t=1}^{T} \frac{\partial \log l}{\partial b} \right]_{b = \tilde{b}_h}$$

$$= \tilde{b}_h + \left[\sum_t x_t x_t' \left(\frac{\partial f_t}{\partial z} \right)^2 \frac{\partial C_t}{\partial m} \right]^{-1} \left[\sum_t x_t' \frac{\partial f_t}{\partial z} \frac{\partial C_t}{\partial m} (y_t - f(x_t \tilde{b}_h)) \right]$$

where $\partial f_t / \partial z$ and $\partial C_t / \partial m$ are the derivatives of $f(z)$ and $C(m)$ evaluated at $z_t = x_t \tilde{b}_h$ and $m_t = f(x_t \tilde{b}_h)$.

No further computation program is needed to apply this method of scoring since it can easily be achieved by iterative GLS. In effect let us consider the nonlinear heteroscedastic pseudo model defined by

$$y_t = f(x_t b) + u_t, \qquad E u_t = 0, \qquad V u_t = \left(\frac{\partial C}{\partial m} (f(x_t b)) \right)^{-1} \qquad (t = 1, \ldots, T).$$

($V u_t$ is the variance associated with the pseudo distribution function.) $f(x_t b)$ may be expanded in a neighbourhood of $\tilde{b}_h$:

$$y_t \simeq f(x_t \tilde{b}_h) + \frac{\partial f}{\partial z} (x_t \tilde{b}_h) \cdot x_t (b - \tilde{b}_h) + u_t .$$

This expression can also be written as

$$y_t - f(x_t \tilde{b}_h) + \frac{\partial f}{\partial z} (x_t \tilde{b}_h) x_t \tilde{b}_h \simeq \frac{\partial f}{\partial z} (x_t \tilde{b}_h) x_t b + u_t \qquad (t = 1, \ldots, T).$$

If we denote by $\hat{b}_{h+1}$ the GLS estimate of b obtained from this linear model with covariance matrix

$$\operatorname{diag} \left\{ \left(\frac{\partial C}{\partial m} (x_t \tilde{b}_h) \right)^{-1} \right\},$$

it is straightforward to show that: $\tilde{b}_{h+1} = \hat{b}_{h+1}$.

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms

REFERENCES

[1] BERGE, C.: *Espaces Topologiques, Fonctions Multivoques*. Paris: Dunod, 1965.

[2] BURGUETE, J., R. GALLANT, AND G. SOUZA: "On Unification of the Asymptotic Theory of Nonlinear Econometric Models," *Econometric Reviews*, 1(1982), 151–190.

[3] GALLANT, R., AND A. HOLLY: "Statistical Inference in An Implicit, Nonlinear, Simultaneous Equation Model in the Context of Maximum Likelihood Estimation," *Econometrica*, 48(1980), 697–720.

[4] GOURIEROUX, C., A. MONFORT, AND A. TROGNON: "Estimation and Test in Probit Models with Serial Correlation," CEPREMAP Discussion Paper No. 8220.

[5] ———: "Pseudo Maximum Likelihood Methods: Applications to Poisson Models," *Econometrica*, 52(1984), 701–720.

[6] HOOD, W., AND T. KOOPMANS: "The Estimation of Simultaneous Linear Economic Relationships," in *Studies in Econometric Method*. New Haven: Yale University Press, 1953.

[7] JENNRICH, R.: "Asymptotic Properties of Nonlinear Least Squares Estimators," *The Annals of Mathematical Statistics*, 40(1969), 633–643.

[8] JOHNSON, N., AND S. KOTZ: *Discrete Distributions*. New York: Wiley, 1972.

[9] MALINVAUD, E.: "The Consistency of Nonlinear Regressions," *The Annals of Mathematical Statistics*, 41(1970), 956–969.

[10] ———: *Méthodes Statistiques de l'Econometrie*, 3eme édition. Paris: Dunod, 1978.

[11] MONFORT, A.: *Cours de Statistique Mathématique*. Paris: Economica, 1982.

[12] ROBINSON, P.: "On the Asymptotic Properties of Estimators of Models Containing Limited Dependent Variables," *Econometrica*, 50(1982), 27–41.

[13] SAWA, T.: "Information Criteria for Discriminating Among Alternative Regression Models," *Econometrica*, 46(1978), 1273–1291.

[14] WHITE, H.: "Maximum Likelihood Estimation of Misspecified Models," *Econometrica*, 50(1982), 1–25.

This content downloaded from
183.109.100.217 on Mon, 12 Jan 2026 11:54:52 UTC
All use subject to https://about.jstor.org/terms